SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0611 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 2 **Total Marks:20**

Annexure A

Case Study

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

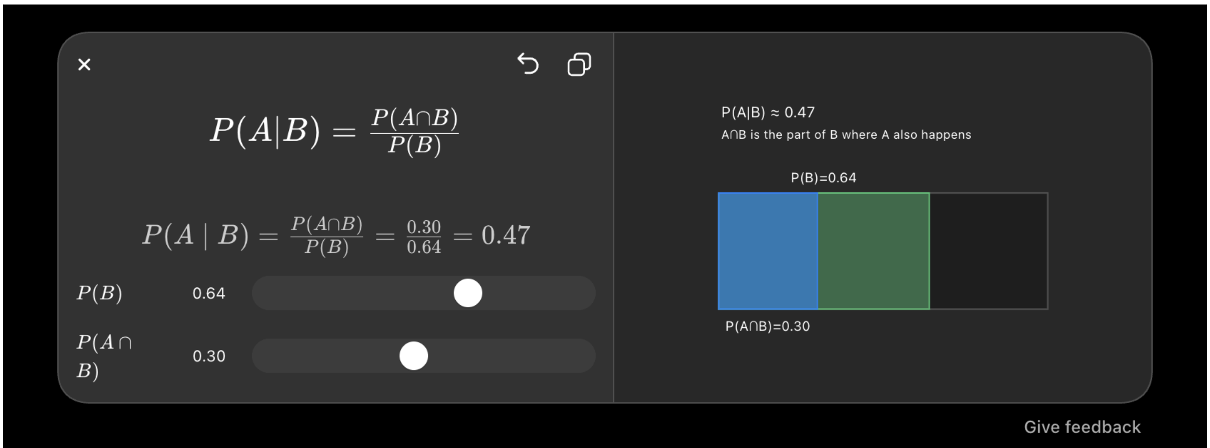
1.Explain how Maximum Likelihood can estimate crop disease probability from agricultural data. (SDG 2 – Zero Hunger) - 50 marks

Analyze how Naïve Bayes can classify recyclable and non-recyclable waste using sensor data. (SDG 11 – Sustainable Cities) - 50 marks

1. Maximum Likelihood Estimation for Crop Disease Probability

Maximum Likelihood Estimation (MLE) is a statistical method used in **Machine Learning** to estimate the most likely values of model parameters from observed data. In agriculture, MLE can be used to estimate the **probability that a crop is affected by a particular disease** based on factors such as temperature, humidity, rainfall, soil moisture, leaf characteristics, and previous disease observations. This can support **SDG 2 – Zero Hunger** by enabling early disease detection and reducing crop losses.

Suppose we have agricultural data containing features such as **temperature, humidity, soil moisture, and leaf symptoms**, along with a target variable indicating whether the crop is diseased (1) or healthy (0). A **Logistic Regression** model can be trained using MLE to predict the probability of disease.



The logistic regression model estimates the probability as:

$$P(Y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\beta_2 X_2+\cdots+\beta_n X_n)}}$$

Here, (Y=1) represents a diseased crop,are agricultural features, and the values are model parameters. MLE determines the values of these parameters that make the observed training data **most likely**.

For example, consider a dataset containing 1,000 crop samples. Each sample records temperature, humidity, soil moisture, and whether the crop developed a disease. The model initially assigns parameter values and calculates the disease probability for every sample. It then uses the **likelihood function** to measure how well those parameters explain the observed outcomes. MLE adjusts the parameters so that the likelihood of the actual observations is maximized. In practice, the **log-likelihood** is usually maximized because it makes the computation easier and more stable.

After training, if the model receives new agricultural conditions such as **high humidity, high temperature, and low soil moisture**, it might produce a disease probability of **0.82 (82%)**. This means the model estimates an 82% probability that the crop has the disease. Farmers or agricultural systems can use this prediction to inspect the crop and take preventive action, such as targeted treatment or improved irrigation.

Therefore, **Maximum Likelihood provides a mathematical foundation for estimating crop disease probabilities from real agricultural data**. When combined with Machine Learning algorithms such as Logistic Regression, it can help identify disease risks early, reduce unnecessary pesticide use, improve crop yield, and support food security. This directly contributes to **SDG 2 – Zero Hunger** by helping farmers protect crops and produce food more efficiently.

2.Naïve Bayes for Classifying Recyclable and Non-Recyclable Waste

Naïve Bayes is a **supervised Machine Learning classification algorithm** based on Bayes' theorem. It can be used in smart waste-management systems to classify waste into categories such as **recyclable** and **non-recyclable** using data collected from sensors. This helps support **SDG 11 – Sustainable Cities and Communities** by improving waste segregation and reducing the amount of recyclable material sent to landfills.

In a smart waste bin, different sensors can collect information about an item, such as **metal detection, weight, moisture level, object size, temperature, and material-related sensor readings**. For example, a metal sensor may indicate that an object is metallic, while a moisture sensor can help distinguish wet organic waste from dry materials.

The Naïve Bayes classifier calculates the probability that the waste belongs to each class based on these observed features:

$$P(Class|Features) \propto P(Class) \times P(Feature_1|Class) \times P(Feature_2|Class) \times \cdots$$

The word **"naïve"** comes from the assumption that the features are conditionally independent of one another given the waste class. Although this assumption is not always completely true, it makes the algorithm simple and computationally efficient.

For example, suppose a waste item has **low moisture, high metal-sensor reading, and medium weight**. The trained model compares these sensor values with previously collected examples of recyclable and non-recyclable waste. It may calculate a probability of **0.90 for recyclable** and **0.10 for non-recyclable**.

The model is trained using a labelled dataset containing sensor readings and their correct waste categories. Its performance can be evaluated using **accuracy, precision, recall, and a confusion matrix**.

Thus, Naïve Bayes provides a **fast, lightweight, and effective approach for sensor-based waste classification**. Integrating it with smart bins and automated sorting systems can improve waste segregation, increase recycling efficiency, reduce landfill waste, and contribute to the development of **cleaner and more sustainable cities**, supporting **SDG 11**.